COMPUTER NETWORKS

Unit-I: Introduction to Computer Networks

Comprehensive Study Notes

1.1 Introduction to Communication Networks

What is a Computer Network?

A **Computer Network** is a collection of interconnected computers and devices that communicate with each other to share resources, exchange data, and enable communication. Networks allow multiple devices to connect and interact, forming the backbone of modern digital communication.

Why Do We Need Computer Networks?

- **Resource Sharing** — Share hardware (printers, scanners) and software (applications, databases)
- **Communication** — Exchange information via email, video conferencing, instant messaging
- **Data Access** — Access remote data from cloud storage and databases
- **Information Sharing** — Share knowledge through websites and collaboration tools
- **Centralized Management** — Manage network resources from a central location
- **Cost Saving** — Reduce expenses by sharing infrastructure and resources

Uses of Computer Networks

Use	Description	Example
Resource Sharing	Share hardware and software	Printers, files, applications
Communication	Exchange information	Email, video conferencing
Data Access	Access remote data	Cloud storage, databases
Information Sharing	Share knowledge	Websites, collaboration tools
Centralized Management	Manage from one place	Network administration
Cost Saving	Reduce expenses	Shared resources

Advantages of Computer Networks

- **Speed** — Fast data transfer and communication
- **Reliability** — Alternative paths and backup options
- **Scalability** — Easy to add new devices and users
- **Security** — Centralized security policies and access control
- **Flexibility** — Access from different locations and devices
- **Cost Efficiency** — Reduced hardware and software costs

1.2 Network Models

Network models define how computers are organized and how they communicate with each other. The three primary models are Peer-to-Peer, Server-Based, and Client-Server.

1.2.1 Peer-to-Peer Network (P2P)

All computers have **equal status**. There is no central server; each computer acts as both client and server.

```
Layout:
┌─────┐     ┌─────┐
│ PC1 │◄───►│ PC2 │
└─────┘     └─────┘
   ▲           ▲
   │           │
   ▼           ▼
┌─────┐     ┌─────┐
│ PC3 │◄───►│ PC4 │
└─────┘     └─────┘
```

Advantages	Disadvantages
Easy to implement	No centralized management
Low cost	Less secure
No dedicated administrator needed	Performance decreases as network grows
Simple for small networks (10–20 computers)	No central backup

1.2.2 Server-Based Network

A **central server** manages the network, provides services, and clients request services from the server.

```
Layout:
        ┌────────┐
        │ Server │
        └───┬────┘
            │
     ┌──────┼──────┐
     │      │      │
  ┌──┴──┐ ┌─┴───┐ ┌─┴───┐
  │ PC1 │ │ PC2 │ │ PC3 │
  └─────┘ └─────┘ └─────┘
```

Advantages	Disadvantages
Centralized management	Expensive (dedicated server)
Better security	Needs skilled administrator
Easy backup	Single point of failure
High performance	Server down = network down

1.2.3 Client-Server Network

Specialized servers handle different functions. Clients request services in a structured, highly scalable environment.

Server Type	Function
File Server	Stores and manages files
Database Server	Manages databases
Web Server	Hosts websites
Print Server	Manages printers
Mail Server	Manages emails

Network Models Comparison

Feature	Peer-to-Peer	Server-Based	Client-Server
Management	Decentralized	Centralized	Centralized
Security	Low	High	High
Cost	Low	High	High
Scalability	Limited	Good	Excellent
Performance	Low	High	High
Size	Small	Medium/Large	Large

1.3 Network Components

Hardware Components

Component	Description	Example
End Devices	Source or destination of data	Computers, printers, phones
Network Devices	Connect and manage network traffic	Routers, switches, hubs
Transmission Media	Carry data signals	Cables, wireless

Software Components

Component	Description	Example
Protocols	Rules for communication	TCP/IP, HTTP, FTP
Operating System	Manage network resources	Windows Server, Linux
Applications	Provide services to users	Web browsers, email clients

1.4 Network Topology

Network Topology is the physical or logical arrangement of devices in a network.

1.4.1 Bus Topology

All devices are connected to a **single cable** (backbone). Data travels in both directions with terminators at both ends.

```
Diagram:
  ┌───────┐ ┌───────┐ ┌───────┐
  │  PC1  │ │  PC2  │ │  PC3  │
  └───┬───┘ └───┬───┘ └───┬───┘
      │         │         │
  ────┴─────────┴─────────┴──── Backbone Cable
      ▲                 ▲
      │                 │
  ┌───┴───┐         ┌───┴───┐
  │Term   │         │Term   │
  └───────┘         └───────┘
```

Advantages	Disadvantages
Simple to install	Single point of failure (cable break = whole network down)
Less cable required	Limited distance
Easy to expand	Hard to troubleshoot
Low cost	Performance degrades with more devices

1.4.2 Ring Topology

Devices are connected in a **circle**. Data travels in one direction and each device acts as a repeater.

```
Diagram:
          ┌──────┐
```

```
     ┌───────► PC1 ◄──────┐
     │       └──────┘     │
     │                    │
     │                    │
┌────┴───┐          ┌─────┴──┐
│  PC4   │          │  PC2   │
└────┬───┘          └────┬───┘
     │                    │
     │       ┌──────┐     │
     └───────► PC3 ◄──────┘
             └──────┘
```

Advantages	Disadvantages
Equal access for all devices	Single point of failure (one device down = whole network down)
No collisions	Hard to troubleshoot
Works well with token passing	Hard to add new devices

1.4.3 Star Topology

All devices are connected to a **central device** (Hub/Switch). The most common topology in modern networks.

```
Diagram:
             ┌──────┐
             │ PC1  │
             └──┬───┘
                │
   ┌──────┐  ┌──┴───┐  ┌──────┐
   │ PC2  ├──┤Hub   ├──┤ PC3  │
   └──────┘  │ /    │  └──────┘
             │Sw    │
   ┌──────┐  └──┬───┘  ┌──────┐
   │ PC4  ├─────┼──────┤ PC5  │
   └──────┘     │     └──────┘
             ┌──┴───┐
             │ PC6  │
             └──────┘
```

Advantages	Disadvantages
Easy to install and manage	Single point of failure (hub down = whole network down)
Easy troubleshooting	More cable required
One device failure does not affect others	Central device is expensive
Scalable	

1.4.4 Mesh Topology

Every device connects to **every other device**. Fully connected and the most reliable topology.

```
Diagram:
     ┌─────┐
     │ PC1 │
     /|\
    / | \
 ┌──┴──┴──┴──┐
 │           │
 │  ┌─────┐  │
 │  │ PC2 │  │
 │  └─────┘  │
 │           │
 │  ┌─────┐  │
 │  │ PC3 │  │
 │  └─────┘  │
 │           │
 └───────────┘

  (Each device connects to all others)
```

Advantages	Disadvantages
High reliability	Very expensive
No single point of failure	Hard to install
Fast communication	Difficult to manage
Secure	Lots of cable required

1.4.5 Tree Topology

A **hierarchical** structure that combines bus and star topologies with parent-child relationships. Also called **Hierarchical Topology**.

```
Diagram:
          ┌─────┐
          │Root │
          └──┬──┘
       ┌─────┼─────┐
       │     │     │
    ┌──┴──┐┌──┴──┐┌──┴──┐
    │Hub A││Hub B││Hub C│
    └──┬──┘└──┬──┘└──┬──┘
    ┌──┼──┐┌──┼──┐┌──┼──┐
    │  │  ││  │  ││  │  │
   PC1 PC2 PC3 PC4 PC5 PC6
```

Advantages	Disadvantages
Scalable	Single point of failure at root
Easy management	Maintenance is complex
Heterogeneous devices supported	Requires skilled staff
Fault isolation	

1.4.6 Hybrid Topology

A combination of **two or more** topologies, customized for specific needs and highly flexible.

Advantages	Disadvantages
Highly flexible	Very expensive
Reliable	Complex to design
Scalable	Hard to manage

Topology Comparison Summary

Topology	Advantages	Disadvantages	Cost	Reliability
Bus	Simple, low cost	Single point failure	Low	Low
Ring	Equal access	Single point failure	Medium	Medium
Star	Easy management	Hub failure affects all	Medium	High
Mesh	Very reliable	Very expensive	Very High	Very High
Tree	Scalable	Root failure critical	High	Medium
Hybrid	Flexible	Complex	Very High	High

1.5 Types of Networks

Networks are classified based on their **geographical coverage** and scale.

1.5.1 Local Area Network (LAN)

Feature	Description
Size	Small (Building, office)
Distance	Up to few kilometers
Speed	Very high (100 Mbps – 10 Gbps)
Ownership	Privately owned
Cost	Low cost
Error Rate	Low

Advantages: High speed, low cost, easy to set up, secure. **Disadvantages:** Limited distance, limited number of devices.

1.5.2 Metropolitan Area Network (MAN)

Feature	Description
Size	City-wide
Distance	5–50 km
Speed	Medium to high
Ownership	Public/Private
Cost	Medium

Advantages: Covers large area, higher speed than WAN, cost-effective for city coverage. **Disadvantages:** Hard to set up, expensive infrastructure, security concerns.

1.5.3 Wide Area Network (WAN)

Feature	Description
Size	Very large (Country/World)
Distance	Thousands of km
Speed	Low to medium
Ownership	Public (ISP)
Cost	Very high

Advantages: Global connectivity, connects LANs and MANs, access to Internet. **Disadvantages:** Slow speed, very expensive, high error rate, security issues.

1.5.4 Personal Area Network (PAN)

A **Personal Area Network (PAN)** is a network for interconnecting devices centered around an individual person's workspace, typically within a range of 10 meters.

Feature	Description
Size	Personal (around a person)
Distance	Up to 10 meters
Speed	Low to medium
Ownership	Personal
Technology	Bluetooth, USB, IR

Examples: Connecting a smartphone to a laptop, wireless headset, smartwatch.

Network Types Comparison

Feature	LAN	MAN	WAN	PAN
Geographic Scope	Building/Campus	City	Country/World	Personal space
Speed	10 Mbps – 10 Gbps	1–100 Mbps	56 Kbps – 10 Mbps	1–24 Mbps
Ownership	Private	Public/Private	Public	Personal
Error Rate	Low	Medium	High	Low
Cost	Low	Medium	Very High	Very Low

1.6 OSI Model (Open Systems Interconnection)

The **OSI Model** is a conceptual framework developed by ISO that standardizes the functions of a communication system into **seven distinct layers**. Each layer provides services to the layer above it and receives services from the layer below it.

```
OSI Model - 7 Layers:

┌──────────────────────────────┐
│ 7. Application Layer         │ <-- User interface
├──────────────────────────────┤
│ 6. Presentation Layer        │ <-- Data translation
├──────────────────────────────┤
│ 5. Session Layer             │ <-- Session management
├──────────────────────────────┤
│ 4. Transport Layer           │ <-- End-to-end delivery
├──────────────────────────────┤
│ 3. Network Layer             │ <-- Routing & addressing
├──────────────────────────────┤
│ 2. Data Link Layer           │ <-- Framing & error control
├──────────────────────────────┤
│ 1. Physical Layer            │ <-- Raw bit transmission
└──────────────────────────────┘

       ▲
       │ Data Flow
       ▼
 Sender ───────────────────► Receiver
```

Layer 1: Physical Layer

Function: Transmits raw **bits** (0s and 1s) over the physical medium. It defines hardware specifications including voltage levels, timing, cable types, connectors, and data rates.

Devices: Hubs, repeaters, modems, cables, connectors.

Layer 2: Data Link Layer

Function: Provides **node-to-node** data transfer and handles error detection/correction at the frame level. It packages raw bits into frames and adds physical addresses (MAC addresses).

Sub-layers: LLC (Logical Link Control) and MAC (Media Access Control).

Devices: Switches, bridges, network interface cards (NICs).

Layer 3: Network Layer

Function: Handles **routing** and **forwarding** of data packets across networks. It determines the best path from source to destination using logical addressing (IP addresses).

Protocols: IP (IPv4, IPv6), ICMP, ARP, RIP, OSPF.

Devices: Routers, layer 3 switches.

Layer 4: Transport Layer

Function: Provides **end-to-end** communication and ensures complete data transfer. It handles segmentation, flow control, error control, and reassembly.

Protocols: TCP (connection-oriented, reliable), UDP (connectionless, faster).

Layer 5: Session Layer

Function: Manages **sessions** or connections between applications. It handles session establishment, maintenance, synchronization, and termination (checkpointing).

Protocols: NetBIOS, RPC, PPTP.

Layer 6: Presentation Layer

Function: Translates between application and network formats. It handles **data translation**, encryption/decryption, compression, and character encoding (e.g., ASCII to EBCDIC).

Protocols: SSL/TLS, JPEG, MPEG, ASCII, EBCDIC.

Layer 7: Application Layer

Function: Provides **network services** directly to end-user applications. It is the closest layer to the user and enables applications to communicate over the network.

Protocols: HTTP, FTP, SMTP, DNS, DHCP, Telnet, SNMP.

OSI Model Data Flow

```
Data Flow in OSI Model:

Sender Side (Encapsulation):
 Application → Presentation → Session → Transport → Network → Data Link → F
     |              |              |           |          |           |
     ▼              ▼              ▼           ▼          ▼           ▼
 [Data]    [Data]    [Data]    [Segments] [Packets]  [Frames]     [Bits]
                                         |
                 Segments + Port #       |    Packets + IP addr  Frames + MAC addr
                                        ▼
                           Physical Medium (Cables/Air)

Receiver Side (Decapsulation - Reverse Order):
 Physical → Data Link → Network → Transport → Session → Presentation → Appl
 Bits        Frames      Packets     Segments     Data       Data             D
```

1.7 TCP/IP Model

The **TCP/IP Model** (Transmission Control Protocol / Internet Protocol) is a practical, protocol-based model developed by the U.S. Department of Defense (DARPA). It has **four layers** and is the foundation of the Internet.

```
TCP/IP Model - 4 Layers:
┌──────────────────────────────────────┐
│ 4. Application Layer                 │
│ (HTTP, FTP, SMTP, DNS, DHCP, Telnet) │
├──────────────────────────────────────┤
│ 3. Transport Layer                   │
│ (TCP, UDP)                           │
├──────────────────────────────────────┤
│ 2. Internet Layer                    │
│ (IP, ICMP, ARP, IGMP)               │
├──────────────────────────────────────┤
│ 1. Network Interface Layer           │
│ (Ethernet, Wi-Fi, PPP, Frame Relay) │
└──────────────────────────────────────┘
```

Layer 1: Network Interface Layer

Function: Combines the functions of OSI Physical and Data Link layers. It handles the physical transmission of data over the network medium and specifies how data is framed and addressed on the local network.

Protocols: Ethernet, Wi-Fi (IEEE 802.11), PPP, Frame Relay, ATM.

Layer 2: Internet Layer

Function: Corresponds to OSI Network layer. Responsible for **logical addressing** (IP addresses), **routing** packets across networks, and handling fragmentation and reassembly.

Protocols: IP (IPv4, IPv6), ICMP, ARP, IGMP, IPSec.

Layer 3: Transport Layer

Function: Corresponds to OSI Transport layer. Provides **end-to-end** communication, segmentation, flow control, and error recovery.

- **TCP** — Connection-oriented, reliable, guaranteed delivery, sequencing
- **UDP** — Connectionless, faster, no guarantee, used for streaming and DNS

Layer 4: Application Layer

Function: Combines OSI Session, Presentation, and Application layers. Provides **high-level protocols** for specific applications and services.

Protocol	Full Name	Port	Use
HTTP	HyperText Transfer Protocol	80	Web browsing
HTTPS	HTTP Secure	443	Secure web browsing
FTP	File Transfer Protocol	20/21	File transfer
SMTP	Simple Mail Transfer Protocol	25	Email sending
DNS	Domain Name System	53	Name resolution
DHCP	Dynamic Host Configuration Protocol	67/68	IP address assignment
Telnet	Telecommunication Network	23	Remote login

1.8 Comparison: OSI Model vs TCP/IP Model

Parameter	OSI Model	TCP/IP Model
Full Name	Open Systems Interconnection	Transmission Control Protocol / Internet Protocol
Developed by	ISO (International Organization for Standardization)	DARPA (U.S. Department of Defense)
Year	1984	1970s
Number of Layers	7 layers	4 layers
Approach	Theoretical / Conceptual	Practical / Protocol-based
Layer Division	Session and Presentation are separate layers	Session & Presentation merged into Application layer
Transport Layer	Connection-oriented only	Both connection-oriented (TCP) and connectionless (UDP)
Network/Internet Layer	Supports multiple protocols	Primarily IP
Protocol Dependency	Protocol independent (generic model)	Tightly coupled with TCP/IP protocols
Reliability	Reliability at Transport layer	Reliability at Transport layer (TCP)
Popularity	Reference/teaching model	Used in real-world Internet and networks

Key Differences Summary

- **OSI** is a *theoretical/reference* model; **TCP/IP** is a *practical/implementation* model.
- **OSI** has 7 layers; **TCP/IP** has 4 layers.

- **OSI** separates Session and Presentation layers; **TCP/IP** merges them into Application layer.
- **OSI** Network layer supports multiple protocols; **TCP/IP** Internet layer primarily uses IP.
- **OSI** Transport layer is connection-oriented; **TCP/IP** supports both TCP and UDP.

Layer Mapping

OSI Model	TCP/IP Model
7. Application	4. Application
6. Presentation	
5. Session	
4. Transport	3. Transport
3. Network	2. Internet
2. Data Link	1. Network Interface
1. Physical	

1.9 Internet

The **Internet** is a global network of interconnected computer networks that uses the TCP/IP protocol suite to communicate.

Brief History of Internet

Year	Event	Description
1969	ARPANET	First network by DARPA
1971	Email	First email sent
1973	TCP/IP	Developed by Vint Cerf and Bob Kahn
1983	TCP/IP adopted	ARPANET uses TCP/IP
1984	DNS	Domain Name System introduced
1989	WWW	World Wide Web by Tim Berners-Lee
1993	Mosaic	First graphical web browser
1995	Internet	Commercialization of Internet
2000+	Modern Internet	Mobile, cloud, IoT

Internet Services

Service	Description
World Wide Web	Websites and web applications
Email	Electronic messaging
Social Media	Social networking
Cloud Computing	Storage and services
Streaming	Video and audio content
IoT	Internet of Things

Protocols and Standards Organizations

Organization	Role
IETF	Internet Engineering Task Force — develops Internet standards
IEEE	Institute of Electrical and Electronics Engineers — networking standards (802.x)
ISO	International Organization for Standardization — OSI model
ANSI	American National Standards Institute
ITU	International Telecommunication Union — telecom standards

1.10 Quick Reference Card

Key Definitions

Term	Definition
Computer Network	Interconnected computers sharing resources
Protocol	Set of rules for communication
Topology	Physical/logical arrangement of network
LAN	Network within a small area
MAN	Network within a city
WAN	Network across large distances
PAN	Personal network around an individual
Internet	Global network of networks

Topology Types Summary

```
Star:   Devices → Central Hub
Ring:   Devices → Circular Connection
Bus:    Devices → Single Cable
Mesh:   Devices → Every Other Device
Tree:   Hierarchical Structure
Hybrid: Combination of Multiple
```

Network Types Summary

```
LAN: Small → Building/Campus → High Speed → Private
MAN: Medium → City → Medium Speed → Public/Private
WAN: Large → Country/World → Low Speed → Public
PAN: Tiny → Personal → Very short range → Personal
```

OSI Mnemonic (Top to Bottom)

```
"All People Seem To Need Data Processing"
Application - Presentation - Session - Transport - Network - Data Link - Physical
```

Important Protocols

```
TCP/IP  → Internet communication
HTTP    → Web browsing
FTP     → File transfer
SMTP    → Email sending
DNS     → Name resolution
DHCP    → IP address assignment
```

Previous Year Questions

2 Marks Short Questions

1. What is a computer network?
2. Define network topology.
3. What are the advantages of computer networks?
4. Differentiate between LAN and WAN.
5. What is a protocol in computer networks?
6. Define peer-to-peer network.
7. What are the main components of a network?
8. What is the Internet?
9. What is the OSI model? Name its layers.
10. What is the TCP/IP model? How many layers does it have?

5 Marks Medium Questions

1. Explain the different types of network topologies.
2. What are the advantages and disadvantages of star topology?
3. Differentiate between LAN, MAN, and WAN.
4. Explain the client-server network model.
5. What is the history of the Internet? Discuss briefly.
6. Explain the functions of each layer of the OSI model.
7. Compare OSI model and TCP/IP model.
8. Explain the four layers of the TCP/IP model with protocols.

12 Marks Long Questions

1. (a) What is a computer network? Explain its need, uses, and advantages.
   (b) Explain the different network types (LAN, MAN, WAN, PAN).
2. (a) Explain the different network models (Peer-to-Peer, Server-Based, Client-Server).
   (b) What is network topology? Explain star, ring, and bus topologies with diagrams.
3. (a) Explain the OSI model in detail with the functions of all seven layers.
   (b) Compare OSI and TCP/IP models.
4. (a) Explain the TCP/IP model with all layers and protocols.
   (b) Discuss the history and evolution of the Internet.
5. (a) What is network topology? Explain all six topologies with their advantages and disadvantages.
   (b) Draw a comparison table of different network types.